# Nikhil Mandayam Adyapak

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**Seeking 2027 Full-Time Roles in ML, ML Systems, Software, MLOps or Data Science**

## Education

**University of Wisconsin-Madison** Sep 2025 - May 2027
Masters in Data Science GPA - 3.66/4.00
**Coursework**: Machine Learning, Foundation Models, Database Design & Implementation, Statistical Models, Statistical Methods, Statistical Inference for Data Science | **Teaching Assistantship**: CS564 Database Management Systems

**PES University - Bengaluru, India** Aug 2019 – May 2023
Bachelor of Technology in Computer Science & Engineering GPA - 3.78/4.00
**Specialization**: Machine Intelligence & Data Science | **Teaching Assistantship**: CS203 Statistics for Data Science
**Coursework**: Algorithms, Data Structures, Data Science, Linear Algebra, Machine Learning, Big Data, Cloud Computing

## Work Experience

**Ciroos** Pleasanton, CA (SF Bay Area)
**AI Intern** May 2026 – Aug 2026

- Productionized a **Graph RAG** context egine on a temporal knowledge-graph (**Graphiti+Neo4j**) sourcing SRE agents incident answers in team chat and investigation history (**87%** accuracy).
- Conducted performance evaluations (**evals**) of AI-driven Site Reliability Engineering (SRE) agents by simulating infrastructure faults across Kubernetes and multi-cloud environments (**AWS, GCP, Azure**).
- Authored and fine-tuned detailed instruction pipelines that guide AI agents to accurately perform Root Cause Analysis (**RCA**) and fault diagnosis across varied system domains.

**Bosch Global Software Technologies (BGSW)** Bengaluru, India
**Senior Software Engineer** – AI Systems for Advance Driver Assisted Systems (ADAS) Aug 2023 – Aug 2025

**MLOps (ML in Operations) & Distributed Cloud Workflows**

- Delivered **MLOps platform** for pedestrian detection (**Azure ML Studio + MLFlow**) with end-to-end **ML lifecycle**; enabled **multi-GPU training** (**3x** faster) and cut GPU costs by **60%**.
- Designed and scaled a **Ray Cluster** based distributed **ML training** system on **Azure Kubernetes Service (AKS)**, reducing retraining cycles from **4 weeks** to **1 week** (**75% faster**) and supporting **30+** teams.
- Automated **ML pipeline** CI/CD workflows on **AKS** (**Terraform, Helm, Argo, GitHub Actions**); integrated **Prometheus** alerting, accelerating experiment setup for **multi-tenant** ADAS teams from **3 days** to **5 minutes**.
- Deployed on-prem **Ollama LLM** with Python FastAPI wrapper, enabling private, team-wide LLM access (**150+ users**), zero external data egress.

**Core Machine Learning & Model Evaluation**

- Enhanced ADAS image dataset resolution by **4x** utilizing **ESRGAN** transfer learning on curated datasets.
- Benchmarked **Occlusion Estimation** models on **nuScenes** dataset by engineering LiDAR-depth evaluation pipelines.

**Large-Scale Image Retrieval & Dataset Generation for Autonomous Driving**

- Architected a containerized **FastAPI** backend for a **60M+ vector image-search engine**, deploying **CLIP embeddings** and **ElasticSearch** to reduce edge-case triage by **95%** (4 hours to 10 mins).
- Engineered **scene-understanding** image retrieval pipeline using foundation models (**Mask2Former, Depth-Anything, OWL-ViT, CLIP**) to generate scene graphs, curating fine-grained datasets for ADAS corner cases.
- Integrated **CLIP embedding** re-use with **80%** cross-version alignment, avoiding full re-indexing of legacy image vectors.

**Machine Learning Operations (MLOps) Intern** Jan 2023 – May 2023 | Jun 2022 – Jul 2022

- Built on-premise **MLOps** pipelines (**DVC**, **MLflow**) and accelerated **multi-GPU** object detection training (YOLOv5, Detectron2) on **Azure ML Studio**, cutting setup time from **1 week** to **1 day** (**85% faster**) and training time by **67%**.
- Configured **Azure DevOps** CI/CD pipelines to automate **Docker** model deployments to **Azure Container Registry**.

## Technical Skills

**Programming languages**: Python, C, C++, Java, SQL, R
**ML & Statistics**: PyTorch, TensorFlow, HuggingFace, Transformers, LLM, CNN, GAN, Scikit-learn, OpenCV, matplotlib
**MLOps & Workflows**: Ray, MLFlow, DVC, Argo Workflows, Azure ML Studio, Prometheus, Grafana
**Cloud**: Azure, AWS, AKS, Docker, Terraform, Azure DevOps, GitHub Actions, Linux
**Databases & Tools**: ElasticSearch, PostgreSQL, MySQL, MongoDB, FastAPI, Flask, Streamlit, GitHub

## Projects and Publications

[1] **Code Runtime Complexity Prediction** - ERCICA (Springer), 2023 | Python, TensorFlow, sklearn, Streamlit, NetworkX
Predicted Big-O runtime complexity of C/Java/Python code using static analysis and ML classification with BiLSTM over Abstract Syntax Trees graph embeddings on IBM CodeNet dataset (**Accuracy: 96%**). **[View Publication]**

[2] **Sentinel-Edge AI** - MadData Hackathon 2026 (Qualcomm Track) | Llama 3.2, Langchain, RAG, FastAPI
Selected as **1 of 10 teams** to build an air-gapped **Edge AI** security auditor on Snapdragon X Elite NPU; engineered a RAG pipeline with ChromaDB to detect legal risks and code vulnerabilities without cloud egress. **[View Project]**

[3] **MLOps POC pipeline for Pedestrian Detection** - 2023 | Python, DVC, MLFlow, DAG, Streamlit, SQLite, PyTorch
Implemented on-premise MLOps starter template with DVC + Detectron2 for end-to-end ML lifecycle.**[View Project]**

[4] **Novel ways of decrypting transposition ciphers**, - IEEE smartgencon, 2022 | Python, Optimization, Natural Language
Introduced cipher-breaking optimization techniques over columnar transposition ciphers. **[View Publication]**